

An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

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1. Executive summary

Floods displace millions of people in India each year, and the limiting factor in a rescue is not reaching survivors but locating them.

We propose three one-metre, 6.4 kg multirotor aircraft that search flooded ground as a coordinated group under a single operator, identify people from the air, report their position to approximately one metre, and deliver a relief kit to each. The system requires no mobile network and no internet connection.

The sizing, error budgets and flight software are complete and reproducible, and coordination has been demonstrated across three simulated autopilots. No aircraft has yet flown, so the figures presented here are calculated; the programme below is ordered to replace them with measurements, beginning with the most consequential.

The request is **INR 7,43,004 across 32 phases**, no phase exceeding INR 30,000, each released on the demonstrated result of the preceding one. The first is INR 26,196.

2. The problem

After a flood, the binding constraint is time-to-locate. Boat teams search slowly, at considerable risk, and cannot observe rooftops. Helicopters are scarce, costly per flight hour, and generally committed to evacuation. Neither covers a large area quickly while reporting precise positions.

Conventional small drones do not close this gap. They require one pilot per aircraft, depend on network infrastructure the disaster has removed, and locate survivors without assisting them.

The more difficult problem is detection itself. A person in floodwater presents only head and shoulders, approximately 40 cm across. Detection models accept a fixed, small input, so the conventional approach reduces the image to fit — and the target reduces with it, to a size at which published results report accuracy between 29 % and 61 %. Selecting a smaller, faster model worsens this, because smaller models accept smaller inputs. Two further constraints are specific to flooding: no existing dataset annotates *people* in flood imagery, and thermal imaging is ineffective in water, where skin temperature equilibrates toward that of the water.

3. Our solution

System architecture. The operator defines a search region of arbitrary shape; the software partitions it into three balanced sub-regions and plans a coverage path over each. This has been verified across thirteen non-convex boundary shapes with no path leaving the region and workload imbalance below 0.02 %. The aircraft then operate autonomously, and the test harness demonstrates structurally that no command is transmitted after the mission begins. Detection and geolocation execute on board. Each aircraft carries four kits and can descend, null its groundspeed over a survivor, and release one. Link loss, low battery and operator abort all converge on a single return-and-land behaviour.

Detection approach. The image is not reduced. Each full-resolution frame is divided into 48 overlapping tiles processed at native resolution, and the survey altitude is 40 m rather than 60 m. Together these place the target at **39 pixels** rather than 13 — nine times the area, and outside the size band in which detector performance degrades sharply. Both changes are required; either alone leaves the target within that band.

The approach requires four times the computation, and this was verified as feasible before adoption: 48 tiles at 2 Hz is 96 inferences per second, against 130–160 measured for the selected accelerator. The additional transects

at the lower altitude cost seconds, and total mission duration is marginally shorter than at 60 m because the climb is reduced.

One item remains open. Confirming a survivor across twelve frames requires approximately 3 Hz, which exceeds the accelerator's capacity. A two-stage approach would recover this, using a low-cost model to eliminate the approximately 87 % of tiles containing only water. It will be evaluated against public data before adoption, since a first-pass filter that discards a tile containing a person fails without indication.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that displays mission state and is architecturally incapable of commanding the aircraft, so autonomy cannot be circumvented in use.
3. Autonomy software — area partitioning, path planning, task allocation and failsafe logic — with an automated test suite.
4. A measured detection accuracy figure, replacing the calculated one.
5. A measured determination of whether the two-stage filter permits the capture rate required for twelve-frame confirmation.
6. A publication: image tiling, aerial person detection and low-power onboard inference have not previously been combined for flood conditions.
7. Source code and results published in either outcome.

5. Benefits

Operational. Three aircraft survey ten hectares in under eight minutes and provide a rescue coordinator with a list of coordinates rather than a video feed requiring continuous attention, and do so without network infrastructure.

Economic. The complete system costs less than one hour of helicopter search time, is transportable by road, and is operated by two people.

Institutional. A calibrated thrust stand, a certified scale, a ground station, a tested autonomy codebase and trained students remain with the department after the project concludes.

Strategic. Indian suppliers are selected wherever they meet the specification, both for supply-chain sovereignty and because domestic lead times are approximately half those of imports, which directly increases available flight-test time.

Transferable. Only the training data is flood-specific. The same system applies to earthquake rubble survey, wildfire perimeter mapping and avalanche search.

6. Resources needed

The programme is divided into **32 phases**. Each funds a single purchase, carries one objective, and is released only when the preceding phase has produced its result. **No phase exceeds INR 30,000**; the largest is INR 29,440 and the first is INR 26,196.

The sequence follows engineering dependency: the motor is measured before eleven more are bought, and one aircraft flies a complete mission before either of the others is begun.

Phase schedule

#	Objective	INR	Released on
1	Establish thrust and mass measurement capability	26,196	Approval
2	Verify static thrust against the 3.18 kgf requirement	5,175	Instruments commissioned
3	Aircraft 1 — autopilot and control link	25,990	Thrust verified
4	Aircraft 1 — onboard compute and inference	21,850	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning	27,892	Compute stack operating
6	Aircraft 1 — sensing and communications	29,440	Position fix acquired
7	Aircraft 1 — airframe fabrication	20,144	Avionics integrated
8	Aircraft 1 — battery pack assembly	29,299	Airframe fabricated
9	Aircraft 1 — drive train and payload mechanism	27,098	Pack bench-discharged
10	Aircraft 1 — propulsion completed	15,525	Drive train installed
11	Charging and manual override for flight test	25,300	Aircraft assembled and weighed
12	Field equipment	17,274	Charging verified
13	Aircraft 2 — autopilot and control link	25,990	Aircraft 1 flew a full mission
14	Aircraft 2 — onboard compute and inference	21,850	Autopilot bench-tested
15	Aircraft 2 — centimetre positioning	27,892	Compute stack operating
16	Aircraft 2 — sensing and communications	29,440	Position fix acquired
17	Aircraft 2 — airframe fabrication	20,144	Avionics integrated
18	Aircraft 2 — battery pack assembly	29,299	Airframe fabricated
19	Aircraft 2 — drive train and payload mechanism	27,098	Pack bench-discharged
20	Aircraft 2 — propulsion completed	20,700	Drive train installed
21	Aircraft 3 — autopilot and control link	25,990	Two aircraft, separation verified
22	Aircraft 3 — onboard compute and inference	21,850	Autopilot bench-tested
23	Aircraft 3 — centimetre positioning	27,892	Compute stack operating
24	Aircraft 3 — sensing and communications	29,440	Position fix acquired
25	Aircraft 3 — airframe fabrication	20,144	Avionics integrated
26	Aircraft 3 — battery pack assembly	29,299	Airframe fabricated
27	Aircraft 3 — drive train and payload mechanism	27,098	Pack bench-discharged
28	Aircraft 3 — propulsion completed	20,700	Drive train installed
29	Correction source for centimetre positioning	27,892	First flight completed
30	Stable, repeatable base mounting	4,600	Base receiver acquired
31	Deployable ground segment	28,750	Base station operating
32	Statutory registration	5,750	Ground segment complete
Total		7,43,004	

Phases are rounded to the rupee, so the column sums to within a few rupees of the total.

Phases 3 and 13 carry the decisions that matter. Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

Each component is a specific model at a current Indian retail price, selected against a quantity the design produces. Where a lower-cost alternative was considered and rejected, the reason is stated.

Measurement instruments — phases 1–2 INR 24,500

Item	Model	INR	Rationale
Calibrated scale	Bench scale, 30 kg, 1 g resolution, certified	12,000	Take-off mass is a hard limit. 1 g on a 6 360 g aircraft is 0.02 %, so mass is tracked component by component. Certification makes the figure defensible.
Thrust stand	20 kg load cell, HX711 amplifier, power meter, frame in-house	8,000	The motors ship without a thrust curve, so the governing quantity of the design is unverified. Measuring thrust and current together gives thrust per watt. Commercial stands start near INR 45,000.
Motor (one)	Tarot TL96020, 5008, 340 KV	4,500	One unit measured before eleven are committed. Requirement is 3.18 kgf at full throttle, hover at half, where efficiency peaks and gust authority remains.

Per aircraft, avionics and sensing — 4 phases INR 85,200

Item	Model	INR	Rationale
Flight controller	Holybro Pixhawk 6C Mini	22,600	Two isolated inertial units, so one sensor failure does not end the flight. Runs ArduPilot, which the simulation testing is built on. F4 boards at INR 6,000 lack memory for the logging and failsafe configuration.
GNSS receiver	RTK-capable, RTCM3, ≤ 3 cm CEP	18,000	The largest single error term. Standard GNSS gives a survivor position to 3.9 m; RTK gives 0.75 m. No other expenditure improves this comparably.
Camera and lens	Arducam IMX477, 6 mm CS, fixed focus	11,100	1.55 μm pixels give 1.03 cm/px at 40 m — 39 pixels on a person in water. Hyperfocal is 4.15 m, so fixed focus is sharp everywhere above 30 m; a motorised module adds a moving part that can drift unnoticed.
AI accelerator	Pi AI HAT+ 26 TOPS (Hailo-8)	11,000	Runs the detector on board. Measured at 130–160 FPS against a 96 requirement. The 13 TOPS variant saves INR 4,000 but reaches only 60–80.
Radio and storage	ExpressLRS 2.4 GHz, firmware 3.5+	8,500	Its native mode carries control and telemetry on one port, which removes the need for a second radio — INR 19,000 per aircraft. The earlier firmware mode consumes the link and would leave telemetry without control.
Onboard computer	Raspberry Pi 5, 8 GB	8,000	Hosts the planner, routing, mesh and delivery logic. The only board in this class exposing PCIe, which the accelerator requires.
Mesh radio	5.8 GHz node and antennas	6,000	Three simultaneous video streams. Offered load is 2.5 Mbps on a 20 Mbps channel; the constraint is link margin at range, not bandwidth.

Per aircraft, airframe and drive — 4 phases INR 74,100

Item	Model	INR	Rationale
Motors	Tarot TL96020, 340 KV, 4 off	18,000	3.18 kgf each at full throttle, giving twice the aircraft weight in thrust with hover at half. 340 KV suits an 18 in propeller at this pack voltage.
Airframe	25 × 23 mm carbon tube, machined clamps, in-house	13,000	Bending stress at design thrust is about 4 % of material capability. Bending is not the failure mode; clamp design and vibration fatigue are, so joints are machined rather than printed.
Battery cells	Molicel P45B 21700, 4500 mAh, 18 off	12,600	Six-series, three-parallel: 292 Wh, covering the mission plus a full second search plus four minutes loiter. Peak is 38.3 A per cell, so a 40 A cell would sit at 96 % of rating.
Pack assembly	6S 60 A BMS, distribution board, regulator	8,500	Cells arrive loose. Covers welding, fusing, balancing and monitoring.
Speed controllers	50–60 A continuous, 6S, 4 off	7,200	Peak demand is 29 A per motor, so these run near half rating and stay within thermal limits. Undersizing here is a known cause of in-flight failure.
Propellers	Tarot 1855 carbon, CW/CCW, 4 off	6,500	18 in carbon matched to motor and pack voltage, at INR 3,250 per pair. The most frequently replaced item in a flight campaign.
Payload release	Four metal-gear servos, mechanical detents	4,500	Detents rather than holding torque, so a power interruption cannot release a kit. Four stations, released individually.
Wiring	10 AWG silicone, XT90-S, JST-GH	2,400	Gauge set by the 115 A peak. The anti-spark connector prevents contact pitting when a 292 Wh pack is connected.
Propeller adapters	6 mm self-tightening hubs, 4 off	1,400	Rotation seats the nut rather than loosening it. A propeller released at operating speed loses the aircraft.

Flight-test support — phases 11–12 INR 36,500

Item	Model	INR	Rationale
Battery chargers	ToolkitRC M6D, 500 W, 25 A, dual, 2 off	11,672	A 13.5 Ah pack needs about 340 W to charge in an hour. The commonly specified HOTA D6 Pro (INR 10,499–INR 13,500) gives 325 W per channel and would run at its limit. The M6D gives 500 W at INR 5,836 and needs only an external supply the department holds. Four channels charge the fleet in parallel.
Consumables	Fasteners, thread-lock, tape, filament	13,000	Consumed continuously in assembly and repair; interruption halts build progress.
Safety-pilot radio	RadioMaster Boxer, ExpressLRS	6,955	Manual override, required every flight. Full-size hall gimbals, as assuming control of a 6.4 kg aircraft under fault conditions demands precise input.
Sun hood and monitor	Fabricated hood, second-hand monitor	1,500	Laptop displays are illegible in direct sun, and an operator who cannot read the display cannot supervise the flight.

Ground segment and statutory — phases 29–32 INR 52,000

Item	Model	INR	Rationale
Field consumables	Cables, mounts, spares, transport protection	25,000	Corrections travel over the existing telemetry link, so no separate base-to-aircraft radio is needed.
Base station	Second GNSS receiver, airborne class	18,000	Centimetre positioning compares the aircraft against a stationary receiver on a known point. Without it, positions degrade from 0.75 m to 3.9 m.
Registration	Statutory, three aircraft	5,000	At 6.36 kg these fall in the Small category of the Drone Rules 2021, so registration precedes any flight.
Survey tripod	Photographic tripod and adapter	4,000	The base needs a stable, repeatable mount rather than survey-grade absolute accuracy, as the corrections are relative.

Structural cost of this arrangement

Thirty-two approvals is a material administrative overhead, and procuring one aircraft at a time requires the same imported order three times, at a schedule cost of approximately eight to twelve weeks. Consolidating the per-aircraft phases into a single order is available if schedule becomes the binding constraint.

The detection evaluation requires no additional procurement: approximately one week on a departmental GPU and 30 GB of storage. The dataset is public and the software is open source.

7. Future work

Near term. Replace each calculated quantity with a measured one: detection accuracy, motor thrust and setup time. Evaluate near-infrared sensing, in which water absorbs strongly and human subjects do not, so a person appears bright against dark water; a camera without its infrared filter costs approximately INR 3,000 and a public dataset exists for evaluation prior to purchase. Beyond flooding, the system applies to earthquake, wildfire and avalanche search.

Longer term: autonomous aircraft under severe power constraint. The central constraint of this work — a capable aircraft deciding for itself within a very small power budget — is a hard requirement in planetary aviation. NASA's *Ingenuity* completed 72 flights on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density. *Dragonfly* launches in 2028 for Titan, where atmospheric density is four times that of Earth at one seventh the gravity.

The difficulty in those missions is not flight but that a radio command requires minutes to arrive: no operator is in the loop, so every decision is made on board. What transfers from this work is the portion that is not flood-specific — onboard survey planning, perception executed to an energy budget rather than an accuracy target, and safe behaviour in the absence of a link. One caution applies: microcontroller-class processors, frequently proposed for such missions, cannot resolve targets of the size considered here, so a planetary configuration would require a different operating point. This is a research question in its own right.

8. Conclusion

The design is complete and explicit as to which quantities are measured and which are calculated. We request the means to complete it in 32 phases, no phase exceeding INR 30,000, so that no expenditure is committed before the preceding phase has produced evidence.

Phases 1 and 2 total INR 31,371 and resolve the question on which the remainder depends: whether the selected motors lift this aircraft. The immediate technical requirement is smaller still — approximately one week of GPU time to evaluate the detection approach against public data.

We would welcome your guidance on two decisions: the operating altitude, which the detection evaluation will inform, and whether to adopt the two-stage detector once it has been measured.